

Robin Wood’s Rock Soundscape

Recording Information

Location	Robin Wood’s Rock, Northumberland Coast National Landscape, north of High Newton by the Sea, UK
Coordinates	55.521° N, –1.609° W
Date & Time	14 January 2025, 14:40 GMT
Recording Equipment	Zoom F8n / Røde NT-SF1 ambisonic microphone
Recording Format	4-channel B-format (AmbiX, ACN/SN3D), 24-bit, 96 kHz

Processing & Render Details

The original 4-channel B-format ambisonic recording was decoded using a first-order UHJ stereo rendering method, optimised for headphone presentation. A 38-second excerpt was processed in REAPER 7.35 using the AmbiX VST plugin suite (Kronlachner, 2014). Spatial decoding used the ambix_binaural_o1 plugin with the icosahedron-3h3v preset. Gain adjustment was applied via JS: Volume Adjustment, followed by limiting via JS: Master Limiter (ceiling –1.0 dBTP). Final metering used Youlean Loudness Meter 2. Output rendered at 24-bit, 44.1 kHz, 2-channel stereo WAV (2067 kbps), 17 April 2025.

Processing Chain

DAW	REAPER 7.35
Ambisonic decoder	AmbiX VST — ambix_binaural_o1, icosahedron-3h3v preset
Gain adjustment	JS: Volume Adjustment
Limiter	JS: Master Limiter (ceiling –1.0 dBTP)
Loudness meter	Youlean Loudness Meter 2
Render date	17 April 2025

Output Specification

Format	WAV, 24-bit PCM
Sample rate	44,100 Hz
Channels	2 (stereo)
Bit rate	2067 kbps
Duration	0:38.000

Loudness Metrics

Parameter	Value
Integrated Loudness (LUFS-I)	–18.3 LUFS
Short-Term Loudness max (LUFS-S)	–13.0 LUFS
Momentary Loudness max (LUFS-M)	–11.3 LUFS
Loudness Range (LRA)	12.9 LU
True Peak	–1.4 dBTP
Clipping	0 occurrences

Loudness normalisation target: $-18\text{ LUFS} \pm 0.5$ (ITU-R BS.1770-4 / AES TD1004.1.15-10).

Rationale

Captures a remote coastal environment with minimal anthropogenic intrusion, featuring coastal wind, distant wave motion, and intermittent seabird calls. Supports analysis of calm, natural soundscapes under the ISO 12913-2 circumplex framework (Pleasant–Uneventful quadrant). Site was initially mislabelled ‘Amble Sea’; geolocation review confirmed the accurate position near Robin Wood’s Rock, north of High Newton by the Sea. Compliant with ITU-R BS.1770-4 and AES TD1004.1.15-10.

REAPER Render Analysis

REAPER Render Thu Apr 17 08:02:56 2025
WAV: 24bit PCM, 44100Hz, 2ch, 2067kbps

File	Length	Peak	Clips	max LUFS-M	max LUFS-S	LUFS-I	LRA
Robin Wood's Rock .wav	0:38.000	-1.4	0	-11.3	-13.0	-18.3	12.9

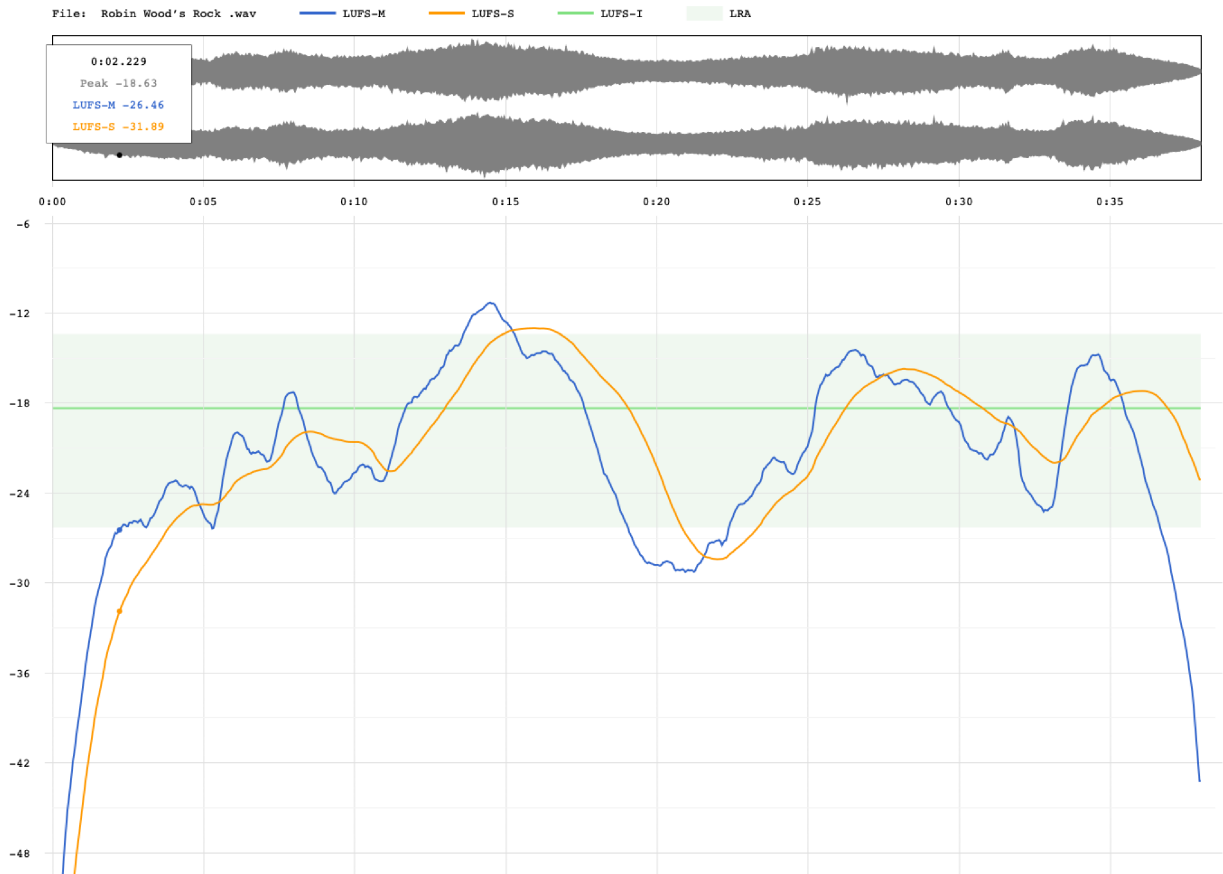


Figure: REAPER loudness render report — Robin Wood's Rock Soundscape. Waveform display (top) and integrated loudness profile (LUFS-M blue, LUFS-S orange, LUFS-I green, LRA shaded region).